ABSTRACT

THE AEGIS–CASCADE TOPOLOGY FAMILY

A Unified Topological Architecture Framework for Neuro-Symbolic Systems

ACT → ACT-D → ACT-E8 → ACT-Ω

ABSTRACT -
This paper introduces the Aegis–Cascade Topology (ACT) family, a unified sequence of computational architectures derived from braid theory, topological invariants, and geometric–combinatorial manifolds. The ACT lineage evolves through four major forms:

ACT — the foundational braid-driven architecture
ACT-D — a topology-driven distilled variant
ACT-E8 — an E8-lattice compressed architecture
ACT-Ω — a sheaf-theoretic manifold integrating E8, Penrose tilings, Tangram polyforms, and Stomachion partitions.

Across all variants, the architecture preserves a core set of invariants: non-commutative cascade behavior, far-commuting side-channels, global writhe, and orientation. These invariants guarantee that distillation and compression preserve the model's function class.

The ACT family demonstrates that topology can serve as a complete design language for neural architectures, model compression, routing, and hardware-agnostic execution.

Introduction

1. Introduction

Modern neural architectures are typically described in terms of layers, tensors, and parameter counts. This paper proposes a fundamentally different paradigm: treating computation as topology.

Two principles motivate this shift:
Braid theory provides a natural language for describing sequential and parallel interactions.Geometric manifolds (E8, Penrose, Tangram, Stomachion) provide structured spaces for routing, compression, and representation.

The ACT family emerges from the observation that: “The compiler searches for mathematically redundant knots during this continuous flow map.”

This insight allows us to treat:redundant parameters → Reidemeister loopsparallel heads → far-commuting strandssequential dependencies → non-commutative braid generatorsmodel compression → topological simplificationThe ACT lineage demonstrates how these principles can be applied systematically.

Mathematical Preliminaries

2. Mathematical Preliminaries

2.1 Braid Groups

The ACT family is grounded in the Artin braid group \(B_n\), with generators:

\[
\sigma_i : \text{cross strand } i \text{ over } i+1
\]

and relations:

- Non-commutativity: \(\sigmai \sigma{i+1} \neq \sigma{i+1} \sigmai\)
- Far-commutativity: \(\sigmai \sigmaj = \sigmaj \sigmai\) for \(|i-j| \ge 2\)
- Reidemeister II: \(\sigmai \sigmai^{-1} = e\)

2.2 Invariants

Across all ACT variants:

- Writhe: \(w = +1\)
- Orientation: preserved
- Function class: preserved
- Cascade Core: preserved
- Aegis Channel: preserved

These invariants ensure zero-loss distillation.

Aegis Cascade Topology

3. ACT — The Aegis–Cascade Topology

3.1 BraidC Specification

```
BRAID 8;
TWIST 2;
TWIST 3;
INVERT 5;
POLYTOPE 24-CELL;
ENTANGLE;
COLLAPSE;
```

3.2 Architecture

- Cascade Core: $\sigma2 \cdot \sigma3$
- Aegis Channel: σ_5^{-1}
- Projection: 24-cell polytope
- Resolution: ENTANGLE → COLLAPSE

3.3 Properties

- Sequential non-commutative core
- Isolated inverse side-channel
- Peripheral bypass strands
- Hardware-agnostic execution

Aegis Cascade Distilled

4. ACT-D — Distilled Aegis–Cascade Topology

4.1 Distillation Rules

- Remove redundant loops: $\sigma3 \sigma3^{-1} \to e$
- Remove dead bypasses: $\sigma7 \sigma7^{-1} \to e$
- Preserve invariants

4.2 Resulting Braid

$$
\beta{\text{ACT-D}} = \sigma1 \cdot \sigma2 \cdot \sigma4^{-1}
$$

4.3 Architecture

- Minimal Cascade Core
- Minimal Aegis Channel
- Simplified projection
- No bypasses

4.4 Benefits

- Lower memory
- Lower latency
- Same function class

ACT-E8

5. ACT-E8 — E8 Cascade Compression

5.1 Key Idea

Use the E8 lattice as the compression manifold:

- Keep high-density regions
- Compress low-density regions
- Snap activations to nearest E8 cell

5.2 Architecture

- E8 global skeleton
- Cascade Core: dense E8 region
- Aegis Channel: orthogonal E8 pocket
- Projection: E8 quantization
- Resolution: ENTANGLE/COLLAPSE

5.3 Benefits

- Structured quantization
- Reduced dimensionality
- Hardware efficiency

ACT-Ω — Omega Architecture

6. ACT-Ω — Omega Architecture

6.1 Overview

ACT-Ω integrates:

- E8 lattice — global symmetry
- Penrose tilings — local aperiodic compute fields
- Tangram polyforms — atomic compute primitives
- Stomachion partitions — global routing states
- Sheaf theory — global consistency

6.2 Architecture Components

E8 Lattice
Defines global geometry.

Penrose Patches
Each E8 cell contains a Penrose tiling.

Tangram Primitives
Each Penrose tile decomposes into Tangram-like polyforms.

Stomachion Router
536 global routing configurations.

Sheaf Resolver
ENTANGLE → glue local patches
COLLAPSE → produce global section

6.3 Properties

- Non-periodic receptive fields
- Combinatorial routing
- Sheaf-consistent global behavior
- Fully reversible reasoning

Hardware-Agnostic Execution

7. Hardware-Agnostic Execution

CPU
- Cascade Core → SIMD lane swaps
- Aegis Channel → sign-flip lanes
- E8 compression → structured quantization

NPU
- Dense core → clustered tiles
- Aegis → sparse tiles
- E8 → fewer active tiles

Quantum-Style
- σ_2, σ_3 → SWAP/CNOT chains
- σ_5^{-1} → conjugate-phase gate
- E8 → constrained state prep
- Stomachion → measurement routing

Distillation across ACT family

8. Distillation Across the ACT Family

ACT → ACT-D
Reidemeister reduction.

ACT-D → ACT-E8
E8-based compression.

ACT-E8 → ACT-Ω
Geometric–combinatorial expansion.

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9. Conclusion

The ACT family demonstrates that topology is a complete design language for neural architectures. By grounding computation in braid theory, geometric manifolds, and sheaf-theoretic consistency, the ACT lineage provides:

- a unified architecture framework
- a principled distillation method
- a hardware-agnostic execution model
- a reversible reasoning system

ACT-Ω represents the culmination of this lineage: a fully geometric, fully topological, fully consistent computational manifold.